Plan for placing and routing the board

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This is the next step after the sheet was fitted and a new board was created from it. It is a plan, not a layout. Nothing here has been routed.

The generator is not the editor

`tools/build_schematic.py` was the way the sheet was rebuilt, because the posted schematic had names and no symbol drawings, and this machine has no KiCad libraries installed. The script embeds the stock symbols and writes the whole sheet again.

It is not a text editor for the conversation. Running it replaces the schematic, including any move made in KiCad. From here the sheet and the board are edited in KiCad. The script stays in the repository so the rebuild can be repeated, and it should not be run again unless that is the deliberate choice.

The earlier passes also ran ahead without stopping to ask. That was the wrong pace. Each step below stops when the file can be opened and looked at.

The sheet is still A2

The schematic paper is A2, 594 by 420 mm. The symbols are not small. KiCad draws them at a fixed size in millimetres. An A2 page, zoomed to fit a screen, makes those symbols look lost.

A4 is 297 by 210 mm. US Letter is 279 by 216 mm. Changing the paper does not shrink the symbols. It only gives them a smaller page, so the same symbols have to sit closer together or they fall off the sheet. The present arrangement uses most of the A2 page, so it will not fit on A4 until the parts are moved. The nets can stay as they are, because the sheet is joined by labels.

A4 landscape is the one to try. Letter is only worth it if the sheet has to be printed on US paper. If the labels collide on A4, stop at A3 rather than making the text smaller.

Do that move in the schematic editor, then save. Do not do it by running the generator.

What the board is today

`Z80_Simple_Computer.kicad_pcb` is a new KiCad 10 board. The posted file that does not open is still `as-posted/Z80_Simple_Computer.kicad_pcb`.

Updating the new board from the schematic copied the 33 footprints onto it. That is not a placement. The large chips are about 18 mm apart, so the DIP-40 packages sit on top of each other, and the resistors and capacitors are stacked in columns. There is no board outline and there are no tracks.

The drawing sheet inside that file says A4. That is the page the board is plotted on. It is not the size of the board.

What was decided

These were asked from the list above and answered on 2026-11-04. They replace the earlier assumptions.

Choice	Decision
Schematic paper	A4 landscape. Move the symbols in KiCad. Do not run the generator.
Copper layers	Four from the start. Front and back are signals. One inner layer is ground. The other inner layer is +5V.
Board size	Eurocard, 160 by 100 mm, 1.6 mm thick, parts on the top. The outline still waits until the footprints fit inside it.
Connectors	DE9 on a short edge. PIO headers on a long edge. Reset switch on an edge a finger can reach.
How +5V comes in	A screw terminal for +5V and GND, on the edge opposite the DE9. The schematic has no power connector, so this is its own sheet change before routing.
Mounting holes	Four M3 holes, one inside each corner, after the outline exists.
DE9 shell	Leave the shell pad unconnected.
Routing	KiCad’s interactive router, after the placement has been looked at. Not FreeRouting.

How the board should be made

Place the parts in the same groups as the sheet, with room to solder a DIP by hand.

- Z80, its 4 MHz oscillator, and the reset parts.
- RAM, ROM, and the two decode chips.
- PIO and the three headers.
- UART, its 1.8432 MHz oscillator, the MAX232, the charge-pump capacitors, and the DE9.
- One 100 nF capacitor beside the supply pins of each chip. On the sheet those capacitors are drawn in a row. On the board they move next to the package they belong to.

Then see whether that placement fits the 160 by 100 mm Eurocard. If it does not, stop and say so. Do not quietly grow the board.

The copper is four layers:

- Front copper: signals.
- Inner 1: ground, not cut up.
- Inner 2: +5V.
- Back copper: signals.

Through-hole pads get thermal relief so they can be soldered. Check the result with DRC, including the comparison against the schematic. A clean DRC does not mean the board has been built or that it runs.

FreeRouting

FreeRouting is a Java autorouter. It can take a Specctra DSN export from KiCad and write a session file of tracks back. It is not installed here. Java is installed, and it is a very new JDK, which FreeRouting releases often do not run on.

There is no skill in this project that designs a board by running FreeRouting. The layout notes used on the other KiCad work say how to group parts, how to keep a board readable, and how to run DRC. They do not autoroute. Another repository has a short laboratory note for trying FreeRouting on a copy of a board. That is an experiment, and it has not been run on this Z80 board.

Use it only after a placement is frozen, and only on a copy. Import the session, look at it, and run DRC. Do not treat the autorouter’s tracks as accepted because the import succeeded. For this board the first routing attempt should be KiCad’s own interactive router, where a bad track can be seen and ripped up.

Order of work

Each step ends by opening the file. The next step does not start in the same breath.

1. In the schematic editor, set the paper to A4 landscape and move the symbols until they sit inside the frame. Do not change the nets. Do not run `tools/build_schematic.py`.
2. Add the screw terminal for +5V and GND on the sheet, then update the board from the schematic. Look at the new footprint and stop.
3. Move the footprints into the groups above. Leave courtyards apart. Decoupling capacitors sit next to their chips.
4. Draw the 160 by 100 mm outline and the four corner M3 holes.
5. Route the supplies first, then the rest, on the four-layer stack above.
6. Run DRC with the schematic check. Fix what it reports. Export a plot of the copper and look at it.

FreeRouting is not part of this pass. The board is still not a computer someone can power on.